Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/14/2005
PI Name:	Flemings, Merton C. Sc.D.		
Project Title:	The Role of Convection and Growth Competition on Phase Selection in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	617-258-6886
Organization Name:	Massachusetts Institute of Technology		
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City:	Cambridge	**State:**	MA
Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/01/2000	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rathz, Tom	**Contact Phone:**	256 544-1409
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG8-1685		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Undercooling in containerless processing has shown promise in producing metastable phases and materials that exhibit improved chemical homogeneity and ultra-fine grain sizes. Stainless steel alloys require special handling in order to achieve a significant undercooling range. In particular, chromium is particularly susceptible to contamination through reaction with crucible material. When attempting to perform rapid solidification tests, the formation of active heterogeneous nucleation sites limits the undercooling that may be achieved. By using a containerless technique, this contamination is minimized. The focus of this program is to study of the effect of microgravity on metastable solidification. By observing changes in the mechanism for nucleation of the stable phase following primary recalescence from undercooled melts, the role of convection in phase selection may be evaluated. The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural		

	evolution. Experiments focus on the use of convection as a processing control parameter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Completed modelling activity.
Bibliography Type:	Description: (Last Updated: 03/16/2017)
Articles in Peer-reviewed Journals	Hyers RW, Matson DM, Kelton KF, Rogers JR. "Convection in containerless processing." Annals of the New York Academy of Sciences, 2004;1027:474-94. http://dx.doi.org/10.1196/annals.1324.038 , Dec-2004
Articles in Peer-reviewed Journals	Matson DM, Fair DJ, Hyers RW, Rogers JR. "Contrasting electrostatic and electromagnetic levitation experimental results for transformation kinetics of steel alloys." Annals of the New York Academy of Sciences, 2004;1027:435-46. http://dx.doi.org/10.1196/annals.1324.035 , Dec-2004
Dissertations and Theses	Backman D. "Transformation kinetics in steel alloys." BS Thesis, Mechanical Engineering, Tufts University, May 2005. , May-2005
Dissertations and Theses	Mueller A. "Partitioning during rapid solidification of undercooled ternary steel alloys." BS Thesis, Mechanical Engineering, Tufts University, May 2005. , May-2005
Papers from Meeting Proceedings	Hanlon AB, Hyers RW, Matson DM. "Modeling the Effects of Internal Convection on Dendritic Evolution in Stainless Steel Alloys." EPD Congress – 2005. 2005 TMS (the Minerals, Metals & Materials Society) Annual Meeting & Exhibition, San Francisco, CA, February 2005. M.E. Schlesinger, ed. Warrendale, PA : TMS, 2005. p. 501-507. , Feb-2005
Papers from Meeting Proceedings	Venkatesh R, Hyers RW, Matson DM. "The Influence of Internal and External Convection in the Transformation Behavior of Fe-Cr-Ni alloys." EPD Congress – 2004. 133rd Annual Meeting & Exhibition of The Minerals, Metals & Materials Society (TMS), Charlotte, NC, March 2004. M.E. Schlesinger, ed. Warrendale, PA : TMS, 2004. p. 337-344. , Mar-2004
Presentation	D.M. Matson "Enhanced growth by mushy-zone remelting" invited seminar for Toshihiko Koseki, Tokyo University, Tokyo Japan Jul-2005
Presentation	D.M. Matson "Mechanism for transformation of ferrite to austenite in hypoeutectic steel alloys" invited seminar for Kazuhiko Kuribayashi, Dept. of Space Biology and Microgravity Sciences, Japan Aerospace Exploration Agence, JAXA, Sagamihara, Japan Jul-2005